

Hyperloop Development Program CONCEPT

Reporting Period 4: Jul 2022-Dec 2022

22-08-2023 – V3

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

- **Aug:** TransPod unveils its FluxJet and says it could shuttle commuters between Calgary and Edmonton in just 45 minutes: [Link](#)
- **Sep:** Nevomo completes 700-meter-long test track construction for its MagRail system: [Link](#)
- **Sep:** Zeleros partners up with ITP Aero for development of propulsion system: [Link](#)
- **Sep:** "Infrastructure Vision @India 2047" also envisages developing seven hyperloop lines, including two for cargo transport: [Link](#)
- **Sep:** Groningen and Lower Saxony join forces for fostering hyperloop development: [Link](#)
- **Oct:** European Commission announces regulatory framework for hyperloop in 2023 work program
- **Oct:** China successfully tests hyperloop-like train inside a vacuum tube: [Link](#)
- **Nov:** TÜV Süd will certify TUM Hyperloop Demonstrator: [Link](#)
- **Nov:** Posco International and Hardt Hyperloop collaborate for hyperloop development: [Link](#)
- **Nov:** EuroTube and Zikulit develop sustainable concrete for hyperloop tubes: [Link](#)
- **Nov:** Hyperloop plans for European Hyperloop Center start early next year: [Link](#)
- **Dec:** Zeleros secures a stake in a multimillion euro battery project in Spain: [Link](#)
- **Dec:** Tata Steel partners with TuTr to develop hyperloop technology: [Link](#)
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REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Jul 2022-Dec 2022. The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Stage 3 (Summer 22 – Summer 23)
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Detail Design Finished
WG EHC	European Hyperloop Center (EHC) Phase A completed	EHC Delivery and start first tests
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go-ahead build-out of route	3 Preliminary route studies concluded
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	System Concept Design Definition finished
WG Program Management	Establish a connected hyperloop ecosystem	Plan for future funding of the ecosystem, finished spatial planning workshop

HDP has progressed into Stage 3 (June 22 – Summer 23) of a total of four stages. The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered, Yellow the Work Package is on-going.

The EHC Working Group is progressing on the temporary project location in Veendam, and the first partners and suppliers have been contracted for the construction. The Consortium Agreement is signed by all partners. The overall HDP program expenditure is delayed compared to planned, due to delay within WG EHC, which can be seen in the financial section of this report.

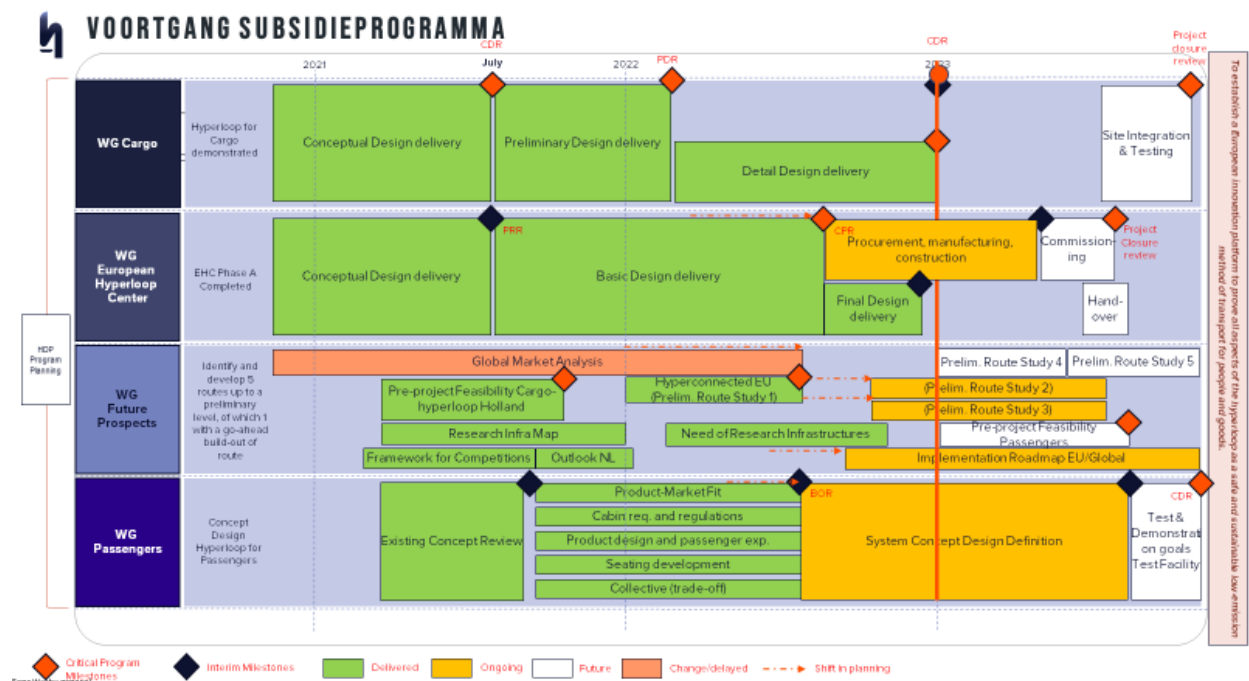


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for open stage 1, stage 2, and started stage 3 is presented below. A number of milestones from stage 1 and 2 have encountered delays, and will be completed in Q3 of 2022.

#	Milestone	Working Group	Status	(Adj.) Date delivery
Open from Stage 1:				
1.9	Final Market opportunity attractiveness	Future Prospects	Ongoing	Q2-23
Stage 2:				
2.1	Basic Design EHC Approved	EHC	Finished	01/09/2022
2.2	Critical Funding and resources are secured for EHC.	Program Management + EHC	Finished	01/12/2022
2.3	EHC Permits approved (Basic Design Approved)	EHC	Finished	01/12/2022
2.4	Partners/suppliers contracted for Manufacturing & Assembly.	EHC	On-going (Delayed)	01/3/2023
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	On-going (Delayed)	Q1-2023
Ongoing Stage 3:				
3.1	Need for Research Infrastructure completed	Future Prospects	Finished	01/11/2022
3.5	Preliminary Route #2	Future Prospects	On-going	Q2-2023
3.6	Preliminary Route #3	Future Prospects	On-going	Q2-2023
3.7	Detail Design Cargo Approved	Cargo	Finished	31/12/2022
3.9	EHC Delivery & Start first tests	EHC	Not Started	Q2/Q3 2023
3.10	Project hand-over EHC	EHC	Not Started	Q2/Q3 2023
3.11	Roadmap EU Finished	Future Prospects	Not Started	Q2/Q3 2023
3.12	Work Package Research Infra finished	Future Prospects	Not Started	Q2/Q3 2023

REPORTING PER WORKING GROUP

WG Cargo

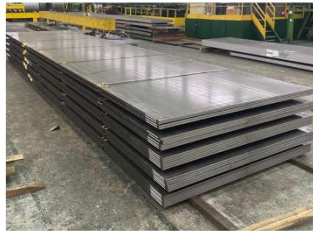
Status: Proceeding according to WG Plan

Within the WG Cargo major process has been made with respect to the detailed design. That of the infrastructure has mostly been completed. During Q4-2022 an intermediate review was organized to evaluate the design status. The detailed design for the tractive and vehicle work packages is not done yet. It is expected to be fully completed in H1-2023. Furthermore, throughout H2-2022 the following highlights have been achieved:

- The steel manufactured by Tata Steel and POSCO for the linear pipe sections was delivered to Noksel (supplier) in Spain and has been used to manufacture the pipes.
- The linear pipes have been shipped to Mercon where they are currently being used in the assembly process.
- The laminated tracks for the magnetic systems have been manufactured and also delivered to Mercon.
- The detail design for all the infrastructure components in the pipes has been completed and written out in various scopes of work.
- Tata Steel and POSCO have started production of steel plate material to be used for the manufacturing of the switch pipe sections.

- Hardt together with the relevant other partners are working on the continuation of the vehicle, infrastructure and the tractive work packages.
- The prototype of the on-board propulsion system was built and commissioning has started. Testing will start in January 2023.

Below a summary in images:



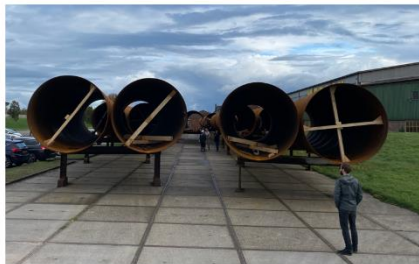
POSCO: Steel plates for hyperloop switch.



Mercon: Steel flanges for linear pipe assembly.



Mercon: Flange assembly jig.



All steel tubes (Ø2.5m) arrived at assembly site.



Electrical steel tracks for lev, guid, and prop arrived at assembly site.

Standardization and regulations

The second half of 2022 sees some great achievements regarding standardization and regulation. The business plan of the JTC20 was finalized and the following standards were submitted for voting:

- Standards Inventory and Roadmap Report
- Hyperloop Transport Services Standard
- Hyperloop Reference Architecture Standard

Next key milestone:

- [HDP-WG-Cargo-3.8] - Detail Design Summary Document Cargo (H2 2023) (Updated)

October, where the assembly has started. Hardt provides Technical Management and coordination based on their design activities within Working Group Cargo.



Welding activities at Mercon

WG Future Prospects

Status: Proceeding according to WG Plan

Work has started on the development of deliverable [FP-D7.3], the EU roadmap. The work has been procured directly by Stichting HDP to be performed by Hypervisory, a Hyperloop Project Advisory Service company. The roadmap covers the market angle, technology angle, regulatory angle, and financial angle. This deliverable is expected to be delivered end of January 2023.

Work has started on the second annual outlook by Berenschot (deliverable [FP-D7.2]). A guideline has been prepared for interviews and interviews have been done with a broad group of stakeholders. The interviews are currently being processed into the deliverable.

A traffic assessment model has been further developed which can be used in modelling in the preliminary studies.

Several studies have been performed, namely Kuala Lumpur – Singapore, Hannover – Bielefeld, Brisbane – Gold Coast, and Ahmedabad – New Delhi, and a corridor in Italy. The results are summarized in the table below:

	Kuala Lumpur – Singapore	Hannover – Bielefeld	Brisbane – Gold Coast	Ahmedabad – New Delhi	RFI Pre-Feasibility Study)
Geography	Malaysia and Singapore	Germany	Australia	India	Italy
Patronage volumes	30 million (passengers per annum in 2050)	214 million (passengers per annum in 2050)	(94 million passengers per annum in 2050)	103 million (passengers per annum in 2069)	11.2 to 34.9 million (passengers per annum in 2050)
Modal share	33%	33%	41%	66%	51%
Preliminary benefit cost ratio (BCR)	1.91	10.41 to 10.90 (includes HYE network effects) < 1.00 without network effects	2.99	1.41 to 2.16	1.01 to 1.41 (includes limited network effects)
Total estimated investment costs (TOTEX)	€7 billion	€8 billion	\$7.3 billion AUD	₹2.5 to ₹3.7 trillion	€29 to 43 billion
Total estimated benefits (in 2022 values)	€26 billion (over 30 years of operation)	€83 to €86 billion (over 30 years of operation)	\$21.7 billion AUD (over 30 years of operation)	₹3.6 to ₹6.2 trillion (over 30 years of operation)	€30 to 61 billion (over 30 years of operation)

The WG has finished main deliverables:

- [FP-D2.8-2-11] Reports on preliminary route study #2 - #5
- [FP-D1.3] Traffic assessment model

Next key milestones:

- FP-D7.3 , Hyperloop roadmap 2023-2030: Europe
- FP-D7.2 , Annual hyperloop outlook: Netherlands